

$$\textcircled{1} \quad \underline{\underline{3 \cdot 2 = 18}}$$

$$x - 3y - z = 12$$

$$12, 0, 0$$

$$x - 3y - z = 0$$

$$[1 \quad -3 \quad -1] \quad x = 0$$

$$A = \begin{bmatrix} \textcircled{1} & \textcircled{-3} & \textcircled{-1} \end{bmatrix}$$

pivot free free

$$\begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

② 2.2 - 24

col space contains $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$N(A)$ contains $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\dim(N(A)) = \underline{\underline{2}}$ at least 2 free vars

$\dim(C(A)) = 2$ at least surely

$N(A) \rightarrow 3$ components then
only 3 columns are there
in A

$$\text{rank} = \dim(C(A)) = 2$$

$$3 - \text{rank} = \dim(N(A))$$

$3 - 2 = 1 \neq \dim(N(A))$

Q 2.2 - 36

$$C = \begin{bmatrix} A \\ B \end{bmatrix}$$

what is relation b/w $N(C)$, $N(A)$, $N(B)$

$$A \rightarrow m \times n$$

$$B \rightarrow p \times n$$

$$C \rightarrow (m+p) \times n$$

$$\begin{bmatrix} A \\ B \end{bmatrix} \quad \begin{bmatrix} \vdots \end{bmatrix}$$

$(m+p) \times n \quad n \times 1$

$$Cx = 0$$

$$\begin{bmatrix} A \\ B \end{bmatrix} x = 0 \rightarrow \begin{bmatrix} A x \\ B x \end{bmatrix} = 0$$

$$N(C) = N(A) \cap N(B)$$

④ 3.2 - 37

Six currents $y_1 \dots y_6$

① $y_1 - y_3 + y_4 = 0$

② $y_1 - y_2 - y_5 = 0$

③ $y_2 - y_3 - y_6 = 0$

④ $y_4 + y_5 + y_6 = 0$

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} y = 0$$

→

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & -1 & 0 \\ 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

→

$$\begin{bmatrix} \textcircled{1} & 0 & -1 & 1 & 0 & 0 \\ 0 & \textcircled{1} & -1 & -1 & -1 & 0 \\ 0 & 0 & 0 & \textcircled{1} & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



$$\begin{bmatrix} \textcircled{1} & 0 & -1 & 0 & -1 & -1 \\ 0 & \textcircled{1} & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & \textcircled{1} & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

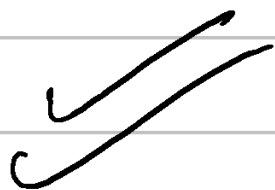
$$\begin{bmatrix} 1 & 0 & -1 & 0 & -1 & -1 \\ 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\underline{\dim(N(A)) = 3}$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$



8) 2.3 - 19

$$A \quad B$$

$n \times n$

$$AB = I$$

$$A = B^{-1}$$

$$B = A^{-1}$$

To prove $\text{rank}(AB) \leq \text{rank}(A)$

$$AB = I$$

Since A is invertible square matrix

$$\text{rank}(I) = n$$

$$\text{rank}(AB) = n$$

$$\text{rank}(AB) \leq \text{rank}(A)$$

$$\boxed{\text{rank}(A) = n}$$

$$\textcircled{6} \quad \underline{\underline{3.3 - 17}}$$

a) j of $B_{m \times n}$ is def
 $A_{p \times m}$

$$B_j = \begin{bmatrix} b_{1j} \\ \vdots \\ b_{mj} \end{bmatrix}$$

$$b_{ij} = c_1 b_{i1} + c_2 b_{i2} \dots c_{j-1} b_{i(j-1)}$$

$\forall i \in [1, m]$

$$(AB)_{:j} = \begin{bmatrix} \text{row vector} \\ [a_i] [b_j] \end{bmatrix}$$

a_i is a common factor.

$$\text{So, } \text{rank}(AB) \leq \text{rank}(B)$$

⑦ 3.3 - 25

$m \times n$ of rank r
 $\downarrow$

$(m \times r) \quad (r \times n)$

$$A = [\text{pivot cols of } A] [\text{first } r \text{ rows}]$$

$$A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 2 & 5 \\ 1 & 3 & 2 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \hat{1} & 0 & 2 & 3 \\ 0 & \hat{1} & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 2 & 4 \\ 1 & 2 & 2 & 5 \\ 1 & 1 & 2 & 6 \end{bmatrix}$$

$$\textcircled{8} \quad \underline{\underline{3.3 - 27}}$$

$$R \rightarrow m \times n \quad (r)$$

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

$$a) \quad I \rightarrow r \times r \quad F \rightarrow (m-r)(n-r)$$

$$b) \quad R \textcircled{B} = I \quad \text{if} \quad r = m$$

$$R \rightarrow \begin{bmatrix} I & F \\ m \times m & \end{bmatrix}$$

$$\begin{bmatrix} I & \vdots & F \\ m \times n \end{bmatrix} B_{n \times m} = \begin{bmatrix} I \\ m \times m \end{bmatrix}$$

$$\begin{bmatrix} I & F \\ m \times m & m \times (n-m) \end{bmatrix}_{m \times n} \begin{bmatrix} I_{m \times m} \\ 0_{(n-m) \times m} \end{bmatrix}$$

$$B = \begin{bmatrix} I_m \\ 0 \end{bmatrix}$$

$$c) \quad CR = I \quad r = n$$

$$C_{n \times m} R_{m \times n} = I_{n \times n}$$

$$C \begin{bmatrix} I_n \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} I_n & 0 \end{bmatrix}$$

$$d) \quad R^T = \begin{bmatrix} I & 0 \\ F & 0 \end{bmatrix}$$

$$= \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$$

$$e) \quad R^T R = \begin{bmatrix} I & 0 \\ F^T & 0 \end{bmatrix} \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} I & F \\ F^T & F^T F \end{bmatrix}$$

$$R^T R = \begin{bmatrix} I & F \\ F^T & F^T F \end{bmatrix}$$

$$R \rightarrow m \times n$$

$$R^T \rightarrow n \times m$$

$$R^T R = (n \times n)$$

$$R^T R \text{'s rref} = \begin{bmatrix} I & P \\ 0 & 0 \end{bmatrix} x_n = 0$$

(9) 3.3 - 28

we can always get to

$$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$$

⑩ $3 \cdot 4 - 13$

a) $\lambda_1 x_p + \lambda_2 x_n$

$\lambda_1 A x_p = \boxed{A, b} \neq \underline{b}$

b) $Ax = b$

No because when there are free variables then we have more than one solution for sure.

c) $\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} x = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

$x_1 = 2$

$x_2 = 0$

$\|x\|^2 = 4$

$x_1 = 1 \quad x_2 = 1 \quad \|x\|^2 = 2$

d) $\{0\}$ is always a solution.

⑪ 3.4 - 25

a) $Ax = b$ no solution

→ no particular solution $[\quad] \checkmark$
for some b

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 4 & 3 \end{bmatrix} x = b$$

$$\hookrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} x = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$r < m$$

b) ∞ solutions for every b .

(free vars) $\checkmark$

$$r < n$$

$$m = r < n \rightarrow \text{no. of sol}^n$$

$\hookrightarrow$ for solvability

c) 1 solution $\rightarrow$ $r = n$

solvable $m > n$

$$r = m = n$$

$$(12) \quad \underline{\underline{3.4 - 28}}$$

$$Ux = 0$$

$$Ux = c$$

$$Rx = 0 \quad Rx = d$$

$$U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix} \quad c = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

$$(1) \quad Ux = 0$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \checkmark$$

$$(2) \quad \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \end{array} \right]$$

$$\hookrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right] \quad x_n \checkmark \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

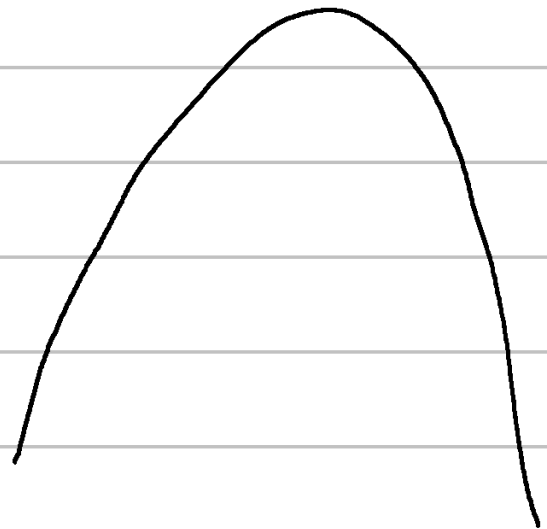
⑬ 3.4 - 35

$K = 9 \times 9$ second diff

$$Kx = b \quad b = \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix}$$

$x =$

$$\begin{bmatrix} 65 \\ 80 \\ 105 \\ 120 \\ 125 \\ \vdots \\ 45 \end{bmatrix}$$



(14) 3.4-36

$Ax = b$ $Cx = b$ same solⁿs
for every b

$$Ax_0 - Cx_0 = 0$$

$$(A - C)x_0 = 0$$

if $x_0 \neq 0$

$$\boxed{A = C}$$